Project Title : Design, Assembly & Flight Control Analysis of a Tandem-Rotor UAV Prototype

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System Architecture : Dual-Rotor Tandem Setup | Custom Flight Controller | Custom Composite Frame

Iterative Design Journey

Version 1 : Proof of Concept

Objective: Baseline thrust, dual-rotor setup and custom dual-axis thrust vectoring gimbal

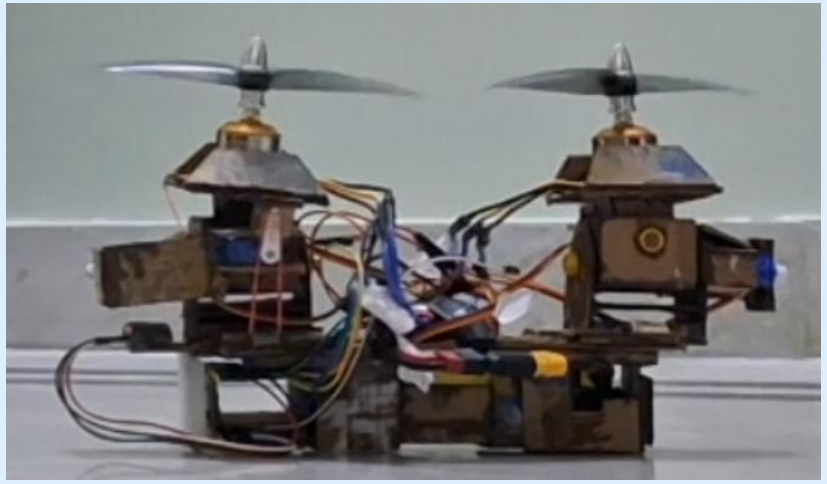
Key Issues: Severe frame flexibility and CG offset



Version 2 : Structural Stability

Objective: Eliminate frame flexibility, improve CG and improve packing efficiency

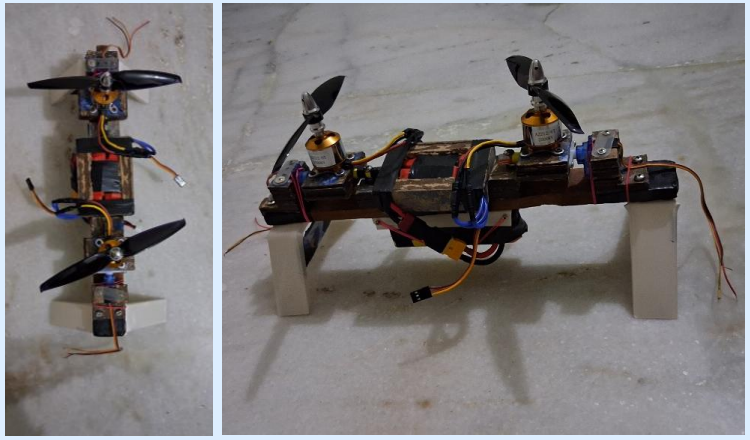
Key Issues: Dynamic coupling and PID instability



Version 3 : Flight Dynamics Integration

Objective: Controlled hover and roll stabilization

Result: Rigid frame, single-axis thrust vectoring gimbal, brief 1-2s hovers, drifted due to ground effect



Key Lessons and Thesis Motivation

+ Hardware Insights: Structural rigidity directly impacts IMU signal noise & PID loop stability
+ Control Insights: Untuned cross-coupling between tandem rotors causes immediate pitch instability
+ Thesis Goals: Apply formal System Identification & Robust PID Control to stabilize complex UAVs
  Design, simulation, and hardware-in-the-loop (HIL) implementation of decoupled control allocation algorithms
  Investigation of stability boundaries and failure modes in multi-rotor systems

Flight test video